

Notes: Pindyck (AER, 1988)
Irreversible Investment, Capacity Choice, and the Value of the Firm

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1 Big picture

1.1 Question

How should a firm choose capacity when investment is irreversible and future demand is uncertain?

The standard net present value rule says to invest when the value of a unit of capital is at least as large as its purchase and installation cost. Pindyck argues that this rule is incomplete when investment is irreversible. Installing capital destroys an option: the option to wait and invest later after observing more information about demand. This lost option value is an opportunity cost of investment.

1.2 Core answer

The paper's main answer is:

A firm should add capacity only when the value of the marginal unit of capacity equals the purchase cost plus the opportunity cost of exercising the option to invest.

In symbols, the optimal incremental investment condition is

$$\Delta V(K^*; \theta) = k + \Delta F(K^*; \theta), \quad (1)$$

where $\Delta V(K; \theta)$ is the value of an additional installed unit of capacity, k is the direct cost of buying and installing that unit, and $\Delta F(K; \theta)$ is the value of the option to install that unit later.

1.3 Main takeaways

- Irreversibility makes investment an option-exercise problem, not a simple static NPV problem.
- A unit of installed capacity is valuable partly because it gives the firm an operating option: it can use the unit when demand is high and leave it idle when demand is low.
- The option to wait before installing capacity is even more important. Demand uncertainty raises both the value of installed capacity and the value of waiting, but the waiting option dominates, so optimal capacity falls with uncertainty.
- Firm value has two components: value of installed capacity and value of growth options. In volatile markets, growth options can be a large fraction of firm value.
- Investment can occur in spurts even when capacity is perfectly divisible. This comes from irreversibility and uncertainty, not necessarily from technological lumpiness.
- Conventional measures of long-run marginal cost can understate true marginal cost because they omit the opportunity cost of exercising the investment option.

2 Incremental contribution of the paper

2.1 Relative to standard investment theory

Traditional investment theory often treats the firm as adding capital until the marginal value of capital equals the purchase and installation cost. This paper adds that when capital is irreversible, the relevant cost is not just the purchase price. The firm also gives up the right to wait.

Incremental point: The marginal cost of capacity includes an option value. Therefore, the investment trigger is higher than the ordinary NPV trigger.

2.2 Relative to earlier real-options work

Earlier real-options papers emphasized whether to undertake a discrete project, such as building a plant. Pindyck shifts attention to **capacity choice**: how large a plant should be, how capacity should be expanded over time, and how this affects firm value.

Incremental point: Real-options logic is applied to the marginal unit of capacity. This makes the optimal capacity problem transparent: value the marginal operating option, value the option to install it, and invest until the two are equal at the margin.

2.3 Relative to valuation theory

The paper decomposes firm value into installed-capital value and growth-option value:

$$W = V(K; \theta) + F(K; \theta). \quad (2)$$

This is not only a valuation identity. It gives an economic interpretation of why firms in volatile markets can be worth much more than the value of their capital in place.

Incremental point: The paper shows why growth options may dominate firm value even for price-taking firms in product markets.

2.4 Relative to industry and antitrust analysis

The paper also connects irreversible investment to the measurement of long-run marginal cost. When investment is irreversible, conventional marginal cost omits the opportunity cost of adding capacity. Therefore, conventional comparisons of price and marginal cost may exaggerate markups.

Incremental point: A real-options adjustment is needed to measure marginal cost in volatile industries.

3 Model environment

3.1 Demand, costs, and capacity

The firm faces inverse demand

$$P = \theta(t) - \gamma Q, \quad (3)$$

where $\theta(t)$ is a stochastic demand shifter. A price-taking firm is the special case $\gamma = 0$.

Demand follows a geometric Brownian motion:

$$d\theta = \alpha\theta dt + \sigma\theta dz, \quad (4)$$

where σ measures demand volatility.

The firm has operating cost

$$C(Q) = c_1Q + \frac{1}{2}c_2Q^2, \quad (5)$$

and capacity constraint

$$Q \leq K. \quad (6)$$

Each unit of capacity costs k to install and investment is irreversible: the firm cannot sell capacity back at cost.

3.2 Two kinds of options

The firm holds two related but distinct options:

- **Operating options:** once a unit of capacity is installed, the firm can decide when to use it. It need not produce with the unit when demand is low.
- **Investment options:** before a unit is installed, the firm can wait and decide later whether to buy it. Installing the unit kills this option.

The value of a marginal unit of installed capacity is $\Delta V(K; \theta)$. The value of the option to install the marginal unit later is $\Delta F(K; \theta)$.

3.3 Firm value decomposition

If capital can be installed in small units, the value of the firm can be written as

$$W = \int_0^K \Delta V(v; \theta) dv + \int_K^\infty \Delta F(v; \theta) dv. \quad (7)$$

The first integral is the value of installed capacity. The second integral is the value of future growth options.

The firm chooses K^* to maximize net value, after subtracting the cost of installed capital. The first-order condition is

$$\Delta V(K^*; \theta) = k + \Delta F(K^*; \theta). \quad (8)$$

Interpretation: The firm invests in capacity up to the point where the marginal installed unit is worth exactly its direct cost plus the value of waiting that is destroyed by installing it.

4 Value of an installed marginal unit

4.1 Operating-option value

The incremental profit from an extra unit of capacity is

$$\Delta\pi(K) = \max\{0, \theta - (2\gamma + c_2)K - c_1\}. \quad (9)$$

The max operator is important. The firm uses the marginal unit only when demand is high enough. If demand is too low, the firm leaves the unit idle.

Thus the value of an installed marginal unit, $\Delta V(K; \theta)$, is like a portfolio of call options on future demand. Higher volatility makes these operating options more valuable because the firm benefits from upside demand states while limiting losses in downside states by not using idle capacity.

4.2 Why this does not imply more capacity

A tempting conclusion is that greater volatility should increase optimal capacity because it raises ΔV . Pindyck shows the opposite. Volatility raises the value of the option to wait, ΔF , even more. Therefore the investment threshold rises, and optimal capacity falls.

5 Investment option and optimal capacity

5.1 Option to install capacity

Before the marginal unit is installed, the firm holds an option. If it exercises the option, it pays k and receives an installed unit worth $\Delta V(K; \theta)$. But by exercising, the firm destroys the option to wait, which is worth $\Delta F(K; \theta)$.

The firm therefore invests only if the value of the installed unit is high enough to compensate both costs:

$$\Delta V(K; \theta) - k \geq \Delta F(K; \theta). \quad (10)$$

At the optimum, equality holds for the marginal unit.

5.2 Effect of uncertainty

Higher σ raises the critical demand level needed to justify investment. In the paper's numerical example with $K = 1$, the investment trigger is $\theta^* = 1.5$ when $\sigma = 0$, but rises to 2.45 when $\sigma = 0.2$ and to 3.44 when $\sigma = 0.4$.

Economic message: Uncertainty creates a value of delay. Even if a project has positive conventional NPV, the firm may rationally wait because exercising now destroys a valuable option.

6 Firm value and growth options

6.1 Installed capital versus growth options

Firm value is the value of installed capacity plus the value of options to add capacity later. Pindyck's simulations show that growth options can be a very large part of market value.

This is especially true when demand is currently low but volatile. Installed capacity may be small or zero, but the option to expand in the future can still be valuable.

6.2 Investment dynamics

Because investment is irreversible, capacity is nondecreasing. It rises only when demand reaches new sufficiently high levels. If demand later falls, the firm may carry excess capacity and leave some of it idle.

The important point is that investment can be intermittent even when capital is perfectly divisible. Capacity expansion occurs in bursts because the firm waits until demand is high enough to justify killing the option to wait.

7 Tables and figures

7.1 Figures 1 and 2: value of installed marginal capacity

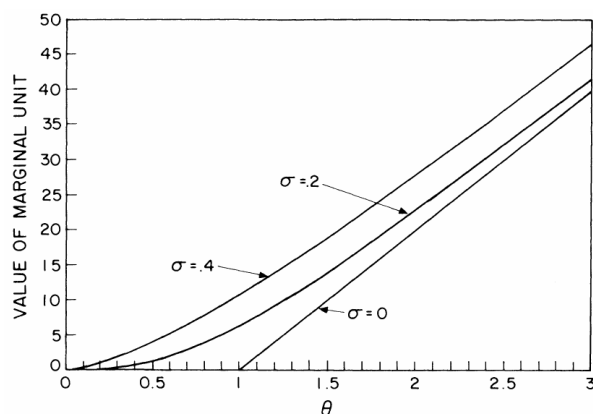


FIGURE 1. THE VALUE OF A MARGINAL UNIT OF CAPACITY, $\Delta V(K)$, AS A FUNCTION OF θ ($K=1$)

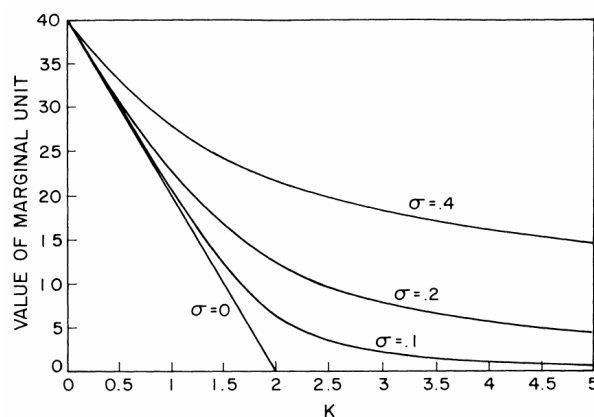


FIGURE 2. THE VALUE OF A MARGINAL UNIT OF CAPACITY, $\Delta V(K)$, AS A FUNCTION OF K ($\theta=2$)

(a) Figure 1. $\Delta V(K)$ as a function of demand, $K = 1$.

(b) Figure 2. $\Delta V(K)$ as a function of capacity, $\theta = 2$.

Original figures extracted from Pindyck (1988).

Read:

- **Figure 1** shows that the value of a marginal unit of capacity increases with demand θ . For a given θ , higher demand volatility σ raises $\Delta V(K)$ because the firm can leave the unit idle in

bad states and use it in good states.

- **Figure 2** shows that $\Delta V(K)$ falls as existing capacity K rises, since the marginal unit is less likely to be used when the firm already has more capacity.
- With uncertainty, $\Delta V(K)$ remains positive over a wider range of K . Even far-out marginal capacity can have value because demand might become high in the future.

Economic message: Installed capacity is valuable not only because it can produce today, but because it gives the firm the option to produce in future high-demand states. This operating option becomes more valuable as demand volatility rises.

7.2 Figures 3 and 4: investment option and optimal capacity

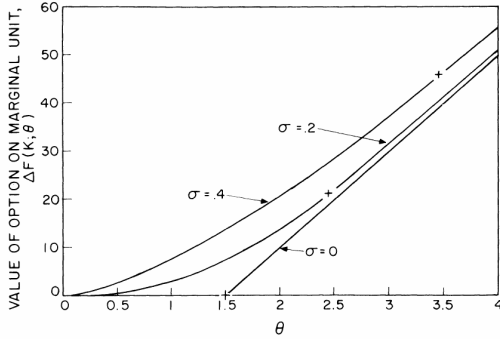


FIGURE 3. THE VALUE OF THE FIRM'S OPTION TO INSTALL A MARGINAL UNIT OF CAPACITY ($K=1$)

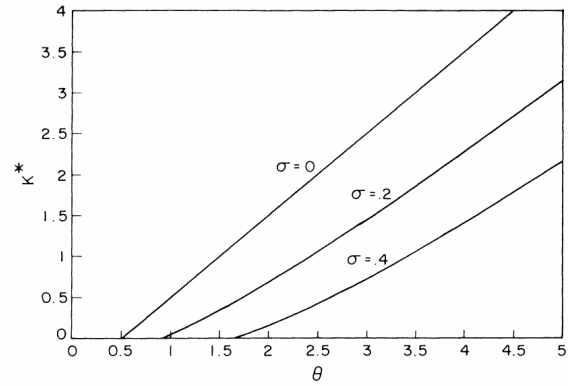


FIGURE 4. OPTIMAL CAPACITY, K^* , AS A FUNCTION OF θ

(a) Figure 3. Value of the option to install a marginal unit.

(b) Figure 4. Optimal capacity K^* as a function of θ .

Original figures extracted from Pindyck (1988).

Read:

- **Figure 3** shows $\Delta F(K)$, the value of the firm's option to install a marginal unit of capacity. The "+" marks the investment trigger θ^* .
- Higher volatility raises the option value and shifts the trigger upward. In the numerical example, the trigger moves from about 1.5 when $\sigma = 0$ to much higher values when $\sigma = 0.2$ or 0.4 .
- **Figure 4** shows the resulting capacity rule. For any given demand level, optimal capacity is lower when demand volatility is higher.

Economic message: Demand uncertainty has two effects: it raises the value of installed capacity, but it raises the value of waiting even more. The net effect is lower capacity and delayed expansion.

7.3 Figure 5 and Table 1: capacity rule and firm value decomposition

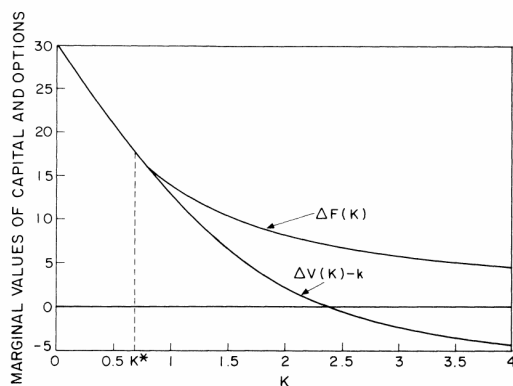


FIGURE 5. THE NET VALUE OF A MARGINAL UNIT OF CAPACITY, AND THE VALUE OF THE OPTION TO INSTALL THE MARGINAL UNIT

(a) Figure 5. Net marginal capacity value and option value.

σ	θ	K^*	$V(K^*)$	$F(K^*)$	Value
0	.5	0	0	0	0
	1	0.5	7.5	0	7.5
	2	1.5	37.5	0	37.5
	3	12.5	87.5	0	87.5
	4	3.5	157.5	0	157.5
.1	.5	0	0	0.4	0.4
	1	0.23	4.3	3.4	7.7
	2	1.00	33.1	9.1	42.2
	3	1.82	80.4	20.3	100.7
	4	2.65	147.3	35.5	182.8
.2	.5	0	0	3.1	3.1
	1	0.04	0.8	13.5	14.3
	2	0.67	24.0	49.2	73.2
	3	1.37	67.1	94.6	161.7
	4	2.09	134.1	143.7	277.8
.4	.5	0	0	25.8	25.8
	1	0	0	69.7	69.7
	2	0.15	5.9	182.6	188.7
	3	0.64	36.2	307.2	343.4
	4	1.22	91.3	427.5	518.8

(b) Table 1. Value of firm.

Original figure and table extracted from Pindyck (1988).

Read:

- **Figure 5** plots $\Delta V(K) - k$ and $\Delta F(K)$. The optimal capacity is the largest K where the net installed value equals the option value.
- If the firm ignored the opportunity cost of investment, it would invest until $\Delta V(K) - k = 0$, which would imply too much capacity.
- In the example, the ordinary NPV logic would suggest capacity of roughly 2.3, but the real-options rule gives only about $K^* = 0.67$.
- **Table 1** decomposes firm value into installed-capital value $V(K^*)$ and growth-option value $F(K^*)$. As σ rises, both total firm value and the share due to growth options rise sharply.
- For $\sigma = 0.2$ and $\theta = 2$, the table reports $K^* = 0.67$, $V(K^*) = 24.0$, $F(K^*) = 49.2$, and total value 73.2. Growth options are already the majority of firm value.

Economic message: The option to wait is not a small correction. It can radically reduce optimal capacity and can account for much of the firm's market value.

7.4 Figure 6: capacity expansion and utilization over time

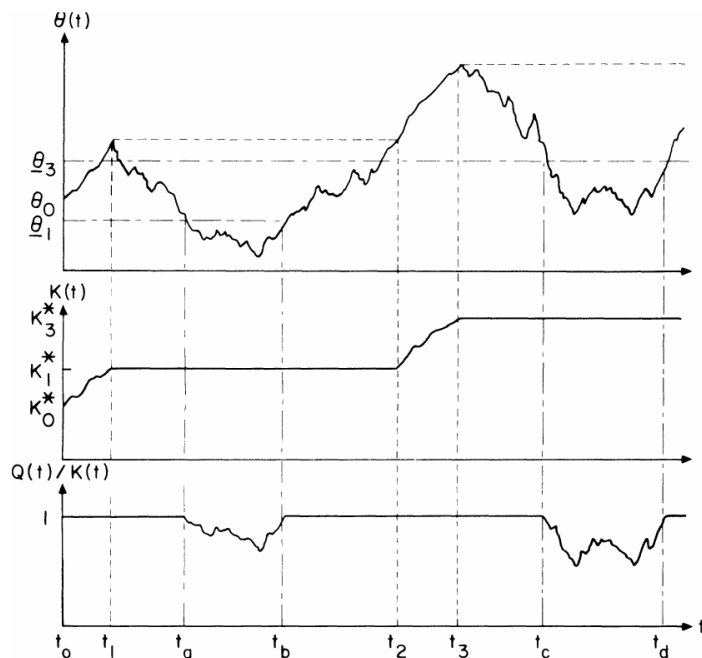


FIGURE 6. REALIZATION OF CAPACITY AND CAPACITY UTILIZATION

Figure 6. Realization of capacity and capacity utilization. Original figure extracted from Pindyck (1988).

Read:

- The top panel shows a sample path of demand $\theta(t)$.
- The middle panel shows capacity $K(t)$. Capacity rises only when demand reaches sufficiently high new levels.
- The bottom panel shows utilization $Q(t)/K(t)$. Capacity is fully used most of the time, but utilization falls below one after sufficiently large demand declines.
- Investment occurs in spurts even though the model allows marginal units of capacity to be installed continuously.

Economic message: Irreversibility makes firms cautious. They hold less buffer capacity against demand declines, expand only when demand is high enough, and then may carry idle capacity after demand falls.

8 Measurement of long-run marginal cost

8.1 Why conventional marginal cost is incomplete

In a conventional long-run analysis, marginal cost includes operating cost plus the purchase cost of capital. With irreversible investment, this is incomplete because the firm gives up the option to wait.

The true full marginal cost has two option adjustments:

- subtract the value of the operating option to leave the marginal unit idle when demand is low;
- add the opportunity cost of exercising the investment option.

In Pindyck's model, the investment-option term dominates, so conventional marginal cost understates true marginal cost.

8.2 Implication

Even in a competitive product market, price can exceed a naive measure of marginal cost if investment is irreversible. The wedge need not imply market power. It may partly reflect the real-options cost of committing capital under uncertainty.

Interpretation: For industries with volatile and unpredictable demand, antitrust or industrial-organization analyses that ignore irreversibility may overstate markups.

9 Why the result matters

9.1 Investment is forward-looking and asymmetric

Firms can choose not to invest, but once they invest they cannot fully undo the action. This asymmetry makes waiting valuable. Investment decisions should therefore be based on threshold rules, not just positive NPV.

9.2 Volatility can reduce real investment

Higher volatility makes operating assets more valuable but can reduce actual installed capacity. This is a key real-options result: uncertainty can depress investment even when projects have positive expected value.

9.3 Growth options are real assets

The paper gives a clean reason why firms can have high market values even with little installed capital. A large part of value can come from the ability to expand later.

9.4 Spurts in investment need not imply indivisibility

The model generates lumpy-looking investment behavior from continuous marginal investment opportunities. The spurt pattern is a consequence of irreversible capacity and the option to wait.

10 Limitations and interpretation

- The model assumes capacity can be added continuously and instantly. Actual investment is often lumpy and may involve construction lags.
- The model has no depreciation. Depreciation would reduce the importance of irreversibility because old capital naturally disappears over time.
- The numerical examples use specific functional forms and parameter values, so the quantitative magnitudes should be interpreted as illustrative.
- The contingent-claims approach assumes that demand risk is spanned by traded assets. If markets are incomplete, a dynamic programming approach can be used, but an arbitrary discount rate is required.
- The model abstracts from strategic interactions among firms, financing frictions, and adjustment costs beyond irreversibility.

Interpretation: The paper should be read as a foundational mechanism paper. Its strongest contribution is conceptual: irreversible investment creates an opportunity cost of capital installation, and this cost changes capacity, utilization, firm value, and marginal-cost measurement.

11 Short summary

Pindyck studies a firm that chooses capacity when investment is irreversible and demand is uncertain. A unit of installed capacity is valuable because the firm can use it in high-demand states and leave it idle in low-demand states. But installing the unit destroys the option to wait and install it later. The correct investment rule is therefore not $\Delta V(K) = k$, but $\Delta V(K) = k + \Delta F(K)$. Demand uncertainty raises the value of installed capacity, but it raises the value of waiting even more. As a result, higher uncertainty lowers optimal capacity and increases the demand threshold required for investment. Firm value can be largely composed of growth options, and investment can occur in spurts even when capacity is continuously divisible. The paper also shows that conventional measures of long-run marginal cost understate true marginal cost when investment is irreversible.

12 Conclusion

The central lesson is that irreversible investment turns capacity choice into a real-options problem. A firm must compare the value of installed capacity with the full cost of installing it, including the opportunity cost of killing the option to wait. This changes how we think about capacity, investment timing, valuation, and marginal cost. In volatile markets, cautious investment and large growth-option values are not anomalies; they are natural consequences of irreversibility and uncertainty.